

2.8 Summary and Notes

2.8.1 Summary

Idea.

1. Expectation of a random matrix is taken entry by entry, and it commutes with constant linear maps, transposes and traces (Proposition 2.1.1).
2. $\text{Cov}(\mathbf{Y}) = \mathbf{E}[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})^T] = \mathbf{E}(\mathbf{Y}\mathbf{Y}^T) - \boldsymbol{\mu}\boldsymbol{\mu}^T$, and $\text{Cov}(\mathbf{U}, \mathbf{V})$ is the rectangular cross-covariance, with $\text{Cov}(\mathbf{V}, \mathbf{U}) = \text{Cov}(\mathbf{U}, \mathbf{V})^T$ (Definition 2.1.2 and Proposition 2.1.2). Uncorrelated does not imply independent (Example (A point on a circle)).
3. Affine maps: $\mathbf{E}(\mathbf{A}\mathbf{Y} + \mathbf{b}) = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}$, $\text{Cov}(\mathbf{A}\mathbf{Y} + \mathbf{b}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T$, $\text{Cov}(\mathbf{A}\mathbf{U}, \mathbf{B}\mathbf{V}) = \mathbf{A}\text{Cov}(\mathbf{U}, \mathbf{V})\mathbf{B}^T$ (Theorem 2.2.1). In particular $\text{Var}(\mathbf{a}^T\mathbf{Y}) = \mathbf{a}^T\boldsymbol{\Sigma}\mathbf{a}$.
4. Covariance matrices are exactly the nonnegative definite matrices. $\boldsymbol{\Sigma}\mathbf{a} = \mathbf{0}$ iff $\mathbf{a}^T\mathbf{Y}$ is constant with probability one, and $\mathbf{Y}$ lives in the flat $\boldsymbol{\mu} + \mathbb{C}(\boldsymbol{\Sigma})$ (Theorem 2.2.3).
5. Whitening by $\mathbf{W}$ with $\mathbf{W}\boldsymbol{\Sigma}\mathbf{W}^T = \mathbf{I}$ is unique up to an orthogonal factor. Every choice gives the same Mahalanobis distance $(\mathbf{Y} - \boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu})$ (Proposition 2.2.4).
6. $\mathbf{E}(\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}) = \text{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}^T\mathbf{A}\boldsymbol{\mu}$ under second moments only (Theorem 2.3.1). Hence $\mathbf{E}(\text{RSS}) = \sigma^2(n - r) + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2$ (Example (Residual sum of squares under a wrong mean)). The variance of a quadratic form needs third and fourth moments (Theorem 2.3.3).
7. Correlation matrices are the nonnegative definite matrices with unit diagonal. They are invariant to rescaling, and their entries must be jointly consistent (Proposition 2.4.1 and Proposition 2.4.2).
8. For partitioned vectors, $\mathbb{C}(\boldsymbol{\Sigma}_{XY}) \subseteq \mathbb{C}(\boldsymbol{\Sigma}_{XX})$. The best linear predictor $\boldsymbol{\mu}_Y + \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}(\mathbf{X} - \boldsymbol{\mu}_X)$ leaves an error uncorrelated with $\mathbf{X}$ with covariance matrix given by the Schur complement $\boldsymbol{\Sigma}_{YY.X}$ (Theorem 2.5.1 and Theorem 2.5.2).
9. Moment generating functions determine distributions and factor exactly under independence (Theorem 2.6.1). Pairwise independence is weaker than mutual independence (Example (Pairwise but not mutually independent)).
10. The sample covariance matrix is unbiased under uncorrelated rows and has rank at most $n - 1$. A tiny eigenvalue of the sample correlation matrix flags a nearly constant combination (Proposition 2.7.1 and Example (Macroeconomic series)).

2.8.2 Notes and sources

The operator calculus. Treating expectation and covariance as operators on random vectors and matrices is standard in linear-model texts. Rencher and Schaalje (2008, chapter 3) develops mean vectors, covariance and correlation matrices, partitioned vectors and linear functions in detail and at a gentle pace. Seber and Lee (2003, §§1.4–1.6) give a compact operator treatment, first and second moments of quadratic forms, and moment generating functions with their use in proving independence. This chapter covers the same ground. Rencher and Schaalje already introduce the standardized distance and the transformation by $\boldsymbol{\Sigma}^{-1/2}$. The chapter adds the characterization of singular covariance matrices through their column space (Theorem 2.2.3),

the non-uniqueness of whitening and the comparison of symmetric and Cholesky whitening, the best linear predictor with generalized inverses, and the sample covariance matrix. Christensen (2020, chapter 1) states the covariance rules briefly in the notation of linear models. For the multivariate setting, Anderson (2003) remains the standard reference.

Mahalanobis distance and whitening. The distance Eq. 2.2.2 is due to Mahalanobis (1936), who introduced it to compare populations described by several correlated measurements. Kessy et al. (2018) compare whitening transformations. They show that the symmetric choice $\Sigma^{-1/2}$ is the one closest to the original data in mean squared error (Exercise (B3)), and set it against Cholesky and principal-component whitening.

Quadratic forms. The trace argument for Theorem 2.3.1 is folklore. Theorem 2.3.3 is stated without proof by Atiqullah (1962), who used it to study how non-normal errors affect the residual variance estimate and the F -test. The kurtosis term depends on $\mathbf{A}$ only through its diagonal. For the residual sum of squares of a correct model it equals $(\mu_4 - 3\sigma^4) \sum_i (1 - h_{ii})^2$, which takes its simplest form, $(\mu_4 - 3\sigma^4)(n - r)^2/n$, when all leverages h_{ii} are equal. These are the quadratically balanced designs of Atiqullah's title. Seber and Lee (2003, theorem 1.6) give a proof by direct expansion. The proof here organizes the same expansion by splitting the form into quadratic, linear and constant parts. Mathai and Provost (1992) is a book-length treatment of moments and distributions of quadratic forms, including general covariance structures.

Correlation, partial covariance and prediction. The feasible range in Proposition 2.4.2 is a special case of the fact that a symmetric matrix is nonnegative definite iff a Schur complement is, when the leading block is positive definite. Horn and Johnson (2013) give the general Schur-complement and Loewner-order results used in Theorem 2.5.2. Stating the best linear predictor with a generalized inverse, so that nothing needs Σ_{XX} to be nonsingular, is in the spirit of Rao (1973), who uses generalized inverses throughout.

Probability background. Uniqueness of moment generating functions and characteristic functions, and the device of Proposition 2.6.2, are proved in Billingsley (1995). The Longley data come from Longley (1967), who used them to test the numerical accuracy of regression programs. Their near-collinearity, visible in Example (Macroeconomic series), is the reason.

2.8.3 References

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